

Olist Dataset Analysis

DSAI 4103 Advanced Business Analytics

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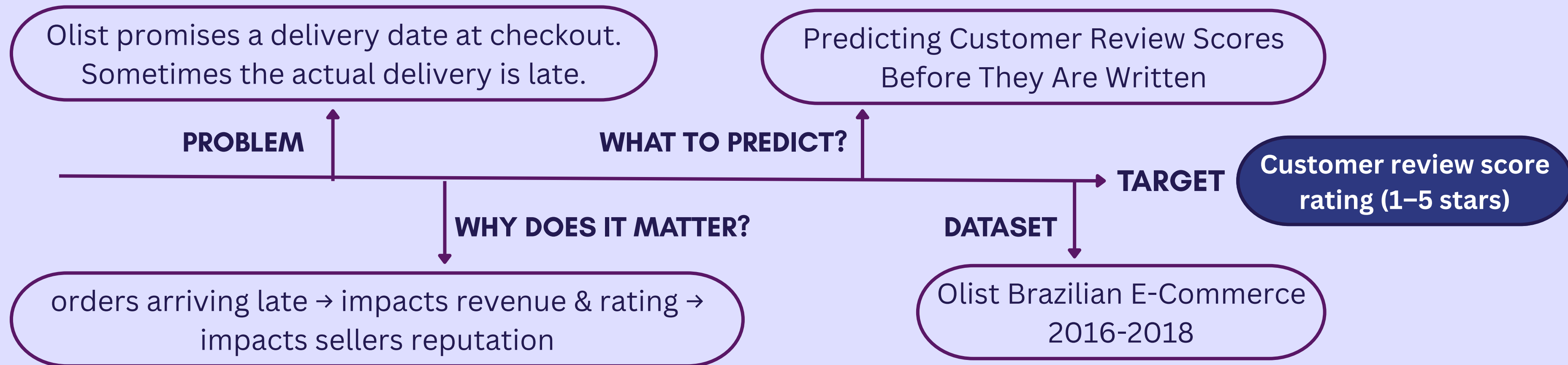
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BUSINESS PROBLEM

predict → **customer review score**



Dataset Understanding

DATASET → Olist Brazilian E-Commerce (2016-2018)

Table	Shape	KEY FEATURES
orders	99441 × 8	order_id, customer_id, status, purchase date, approved date, carrier date, delivery date, estimated date
reviews	99224 × 7	order_id, review_score, comment title, comment message, creation date, answer date
items	112650 × 7	order_id, item_id, product_id, seller_id, price, freight_value
payments	103886 × 5	order_id, payment_type, installments, payment_value
customers	99441 × 5	customer_id, zip_code, city, state
products	32951 × 9	product_id, category, weight, length, height, width, photos
sellers	3095 × 4	seller_id, zip_code, city, state
geoLocation	1000163 × 5	zip_code, lat, lng, city, state
translation	71 × 2	category name Portuguese → English

Data Cleaning & Preprocessing

Join Tables	Central table: orders → everything joins through order_id
Dataset Shape	99,441 total orders
Keeping only delivered	Removed 2,980 (cancelled, invoiced, etc.) → 96,461
Drop null dates	Removed rows missing actual or estimated delivery date → ~96,400
Parse all 5 date columns	Converted string → datetime for arithmetic
Filter payment_type	'not_defined' removed
Median imputation	product_weight, length, height, width nulls → median fill
winsorizing	Values that were too extreme got replaced with the boundary value.
Deduplication	sort by order_id → keep first occurrence

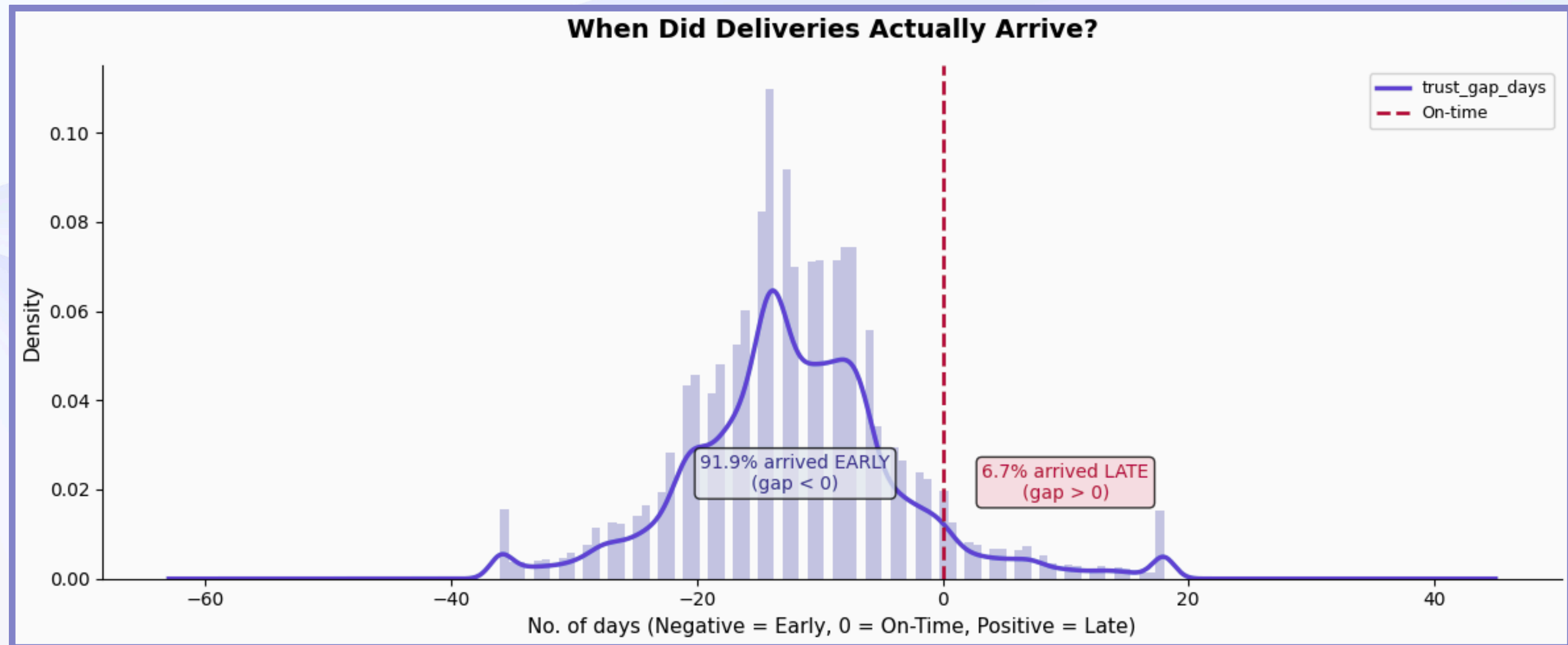
Final Dataset After Cleaning: 96,478 rows × 35 columns

Feature Engineering

trust_gap_days	actual_delivery_date – estimated_delivery_date
actual_delivery_days	order_delivered_customer_date – order_purchase_timestamp
promised_delivery_days	order_estimated_delivery_date – order_purchase_timestamp
shipping_lag_days	order_delivered_carrier_date – order_approved_at
approval_lag_days	order_approved_at – order_purchase_timestamp
freight_ratio	total_freight ÷ (total_price + 0.01)
same_state	1 if seller_state == customer_state else 0 (binary)
state_risk_score	Avg review_score per state → target encoding
category_risk_score	Avg review_score per category → target encoding
seller_avg_score	Mean review_score per seller
seller_late_rate	% of that seller's orders that arrived late
seller_avg_gap	Mean trust_gap_days per seller
seller_order_volume	Count of orders per seller

computed on TRAIN data only to prevent data leakage

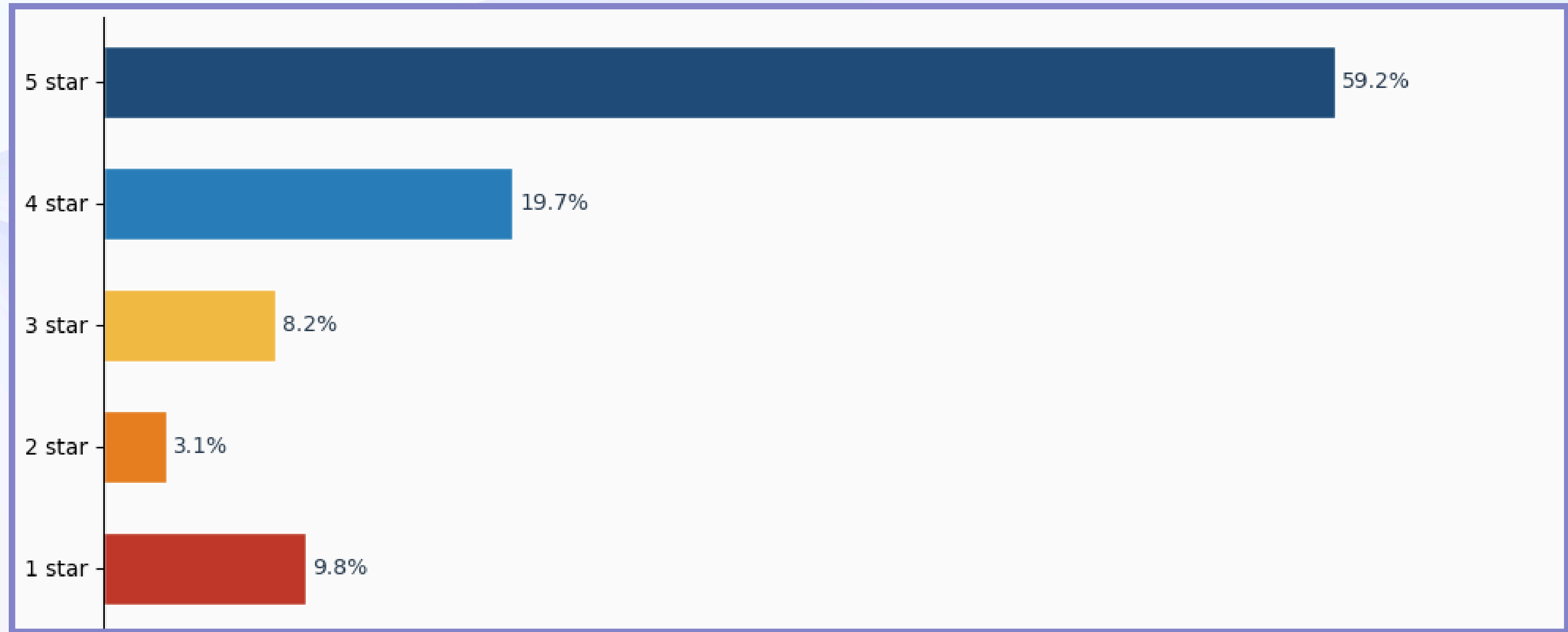
Exploratory Data Analysis



91.9% arrived early → Olist deliberately buffers delivery promises by 11.5 days

Exploratory Data Analysis

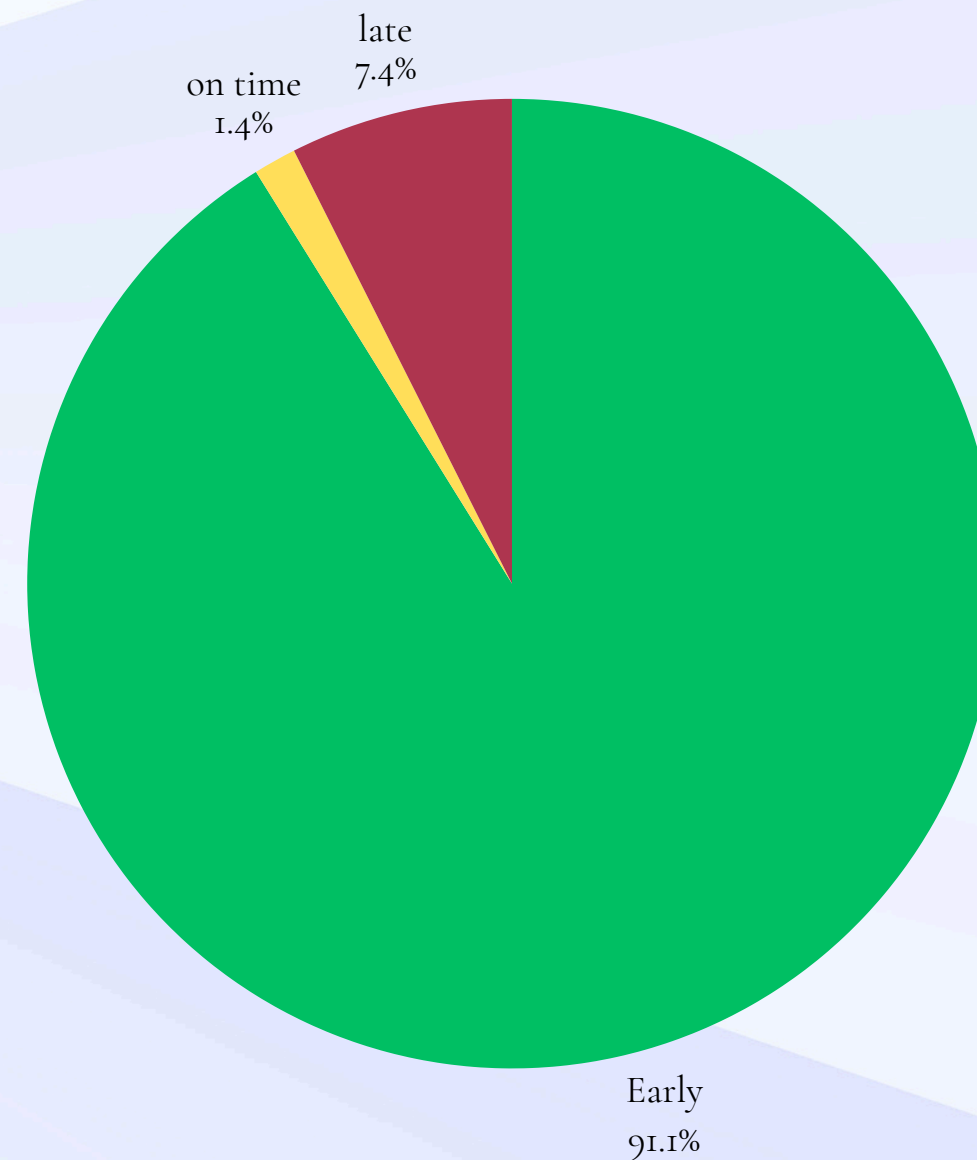
Review Score Distribution



Class imbalance → 5Star → 59.2%

Exploratory Data Analysis

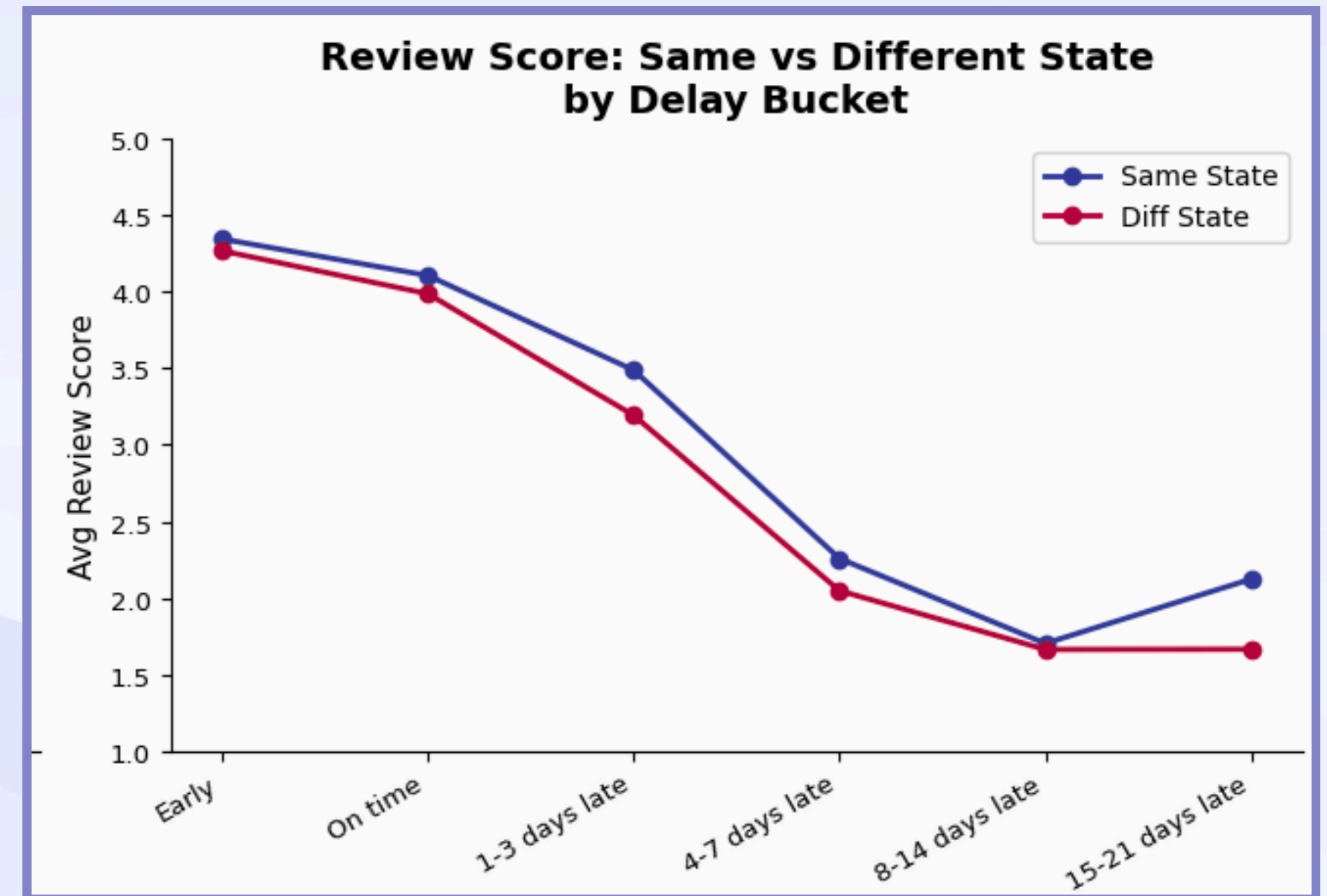
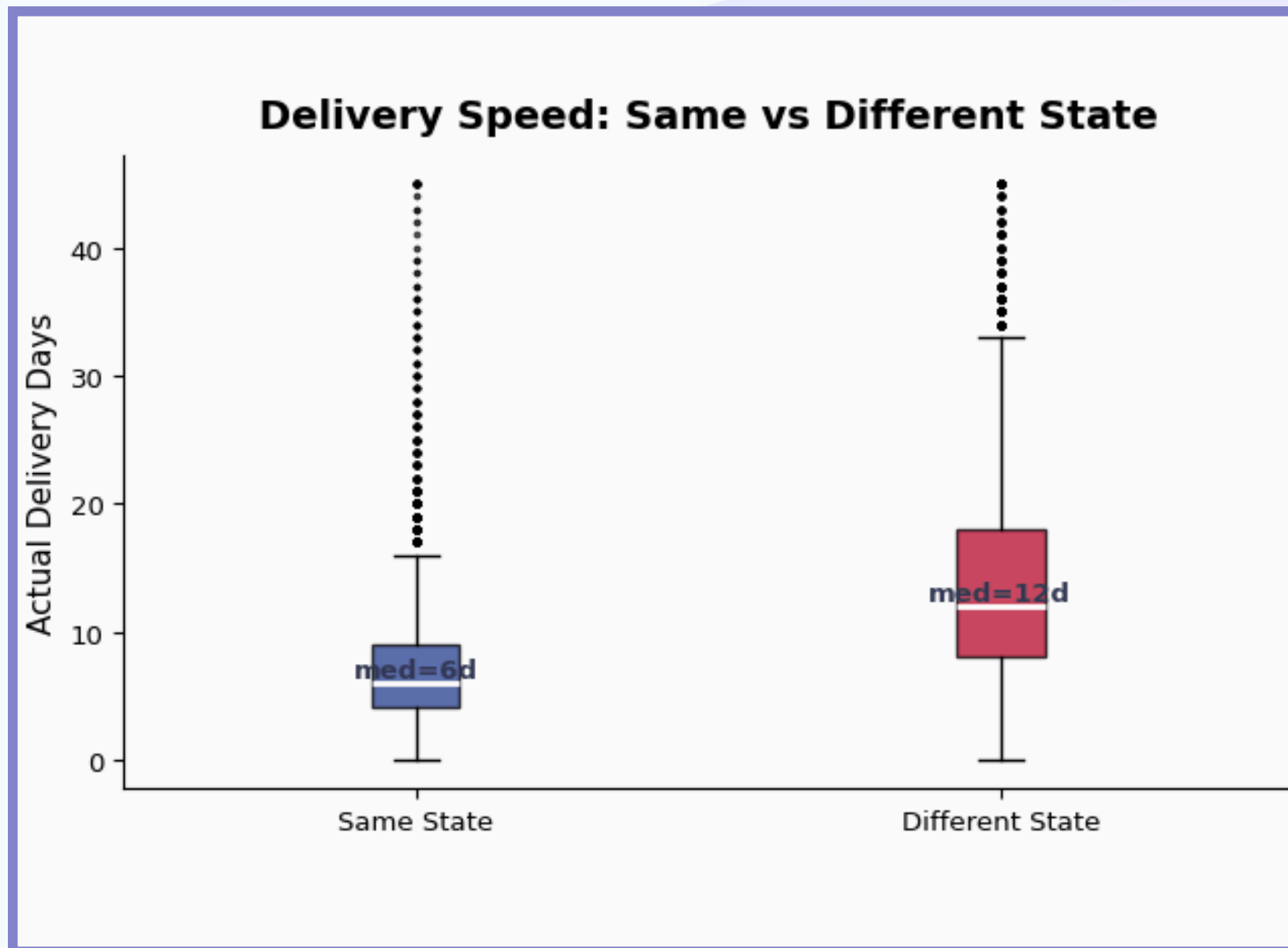
PROMISED VS ACTUAL DELIVERY DATE



Avg buffer days → promised - actual → 11.5 days

Exploratory Data Analysis

Same-State vs Different-State Deliveries



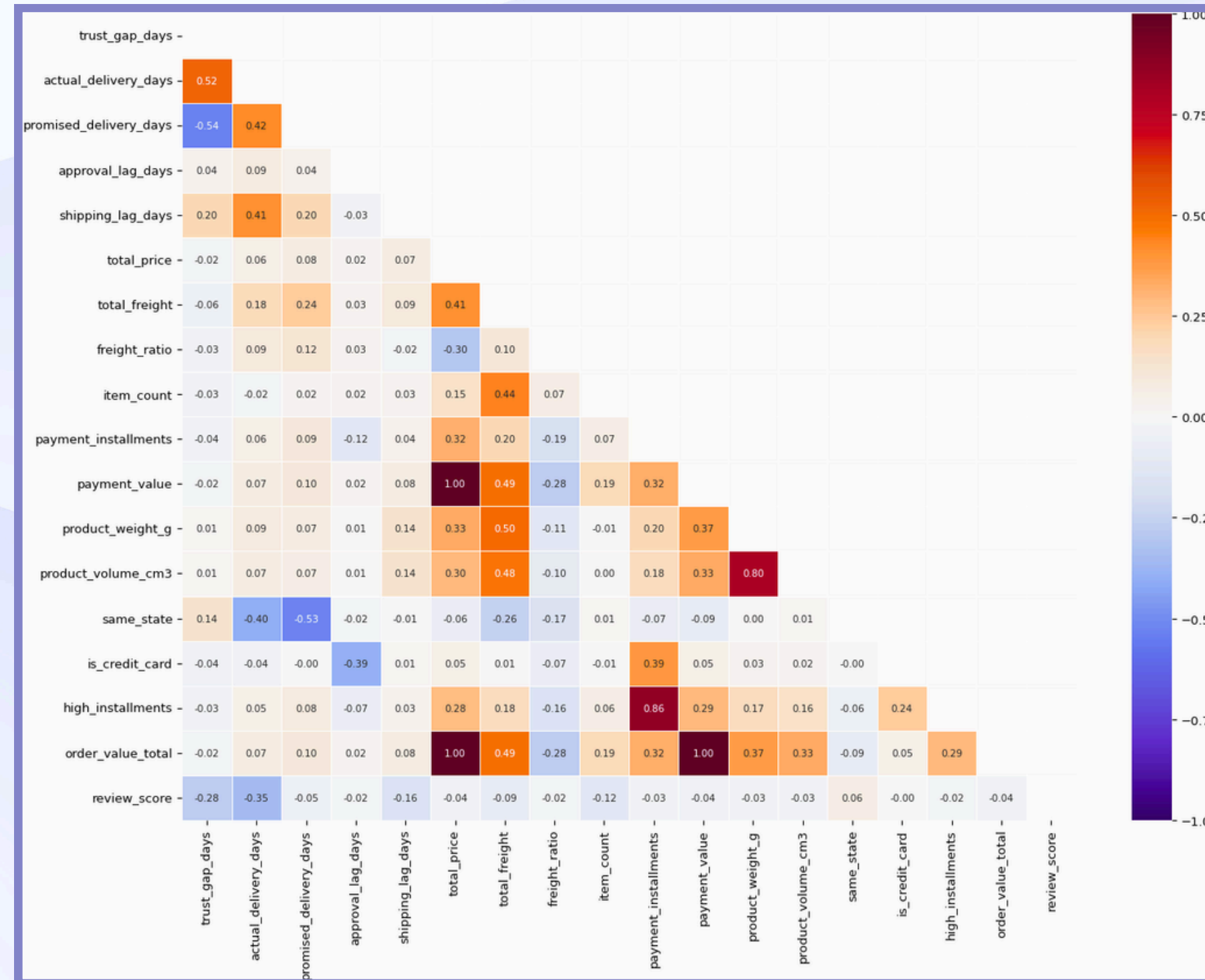
Exploratory Data Analysis

Correlation Matrix

COLLINEAR FEATURE:

*order_value_total,
product_volume_cm3,
high_installment*

DROPPED



Predictive Modeling

25 FEATURES

trust_gap_days, actual_delivery_days,
promised_delivery_days,
shipping_lag_days, approval_lag_days,
total_price, total_freight, freight_ratio,
item_count, payment_installments,
payment_value, pay_credit_card,
pay_debit_card, pay_voucher,
product_weight_g, product_length_cm,
product_height_cm, product_width_cm,
same_state, state_risk_score,
category_risk_score, seller_avg_score,
seller_late_rate, seller_avg_gap,
seller_order_volume

TARGET: REVIEW_SCORE

TEMPORAL SPLITTING

TRAIN	TEST
Oct 2016 → May 2018	May 2018 → Aug 2018
~77,000 orders	~19,162 orders

AUTOML → SELECTED XGBOOST

FINE-TUNING XGBOOST → 500 TREES, DEPTH 6, LEARNING RATE 0.05

WHY XGBOOST?

- HANDLES CLASS IMBALANCE
- WORKS WELL WITH MIXED FEATURE TYPES
- RESISTANT TO OVERFITTING WITH EARLY STOPPING

Model Performance

ACCURACY

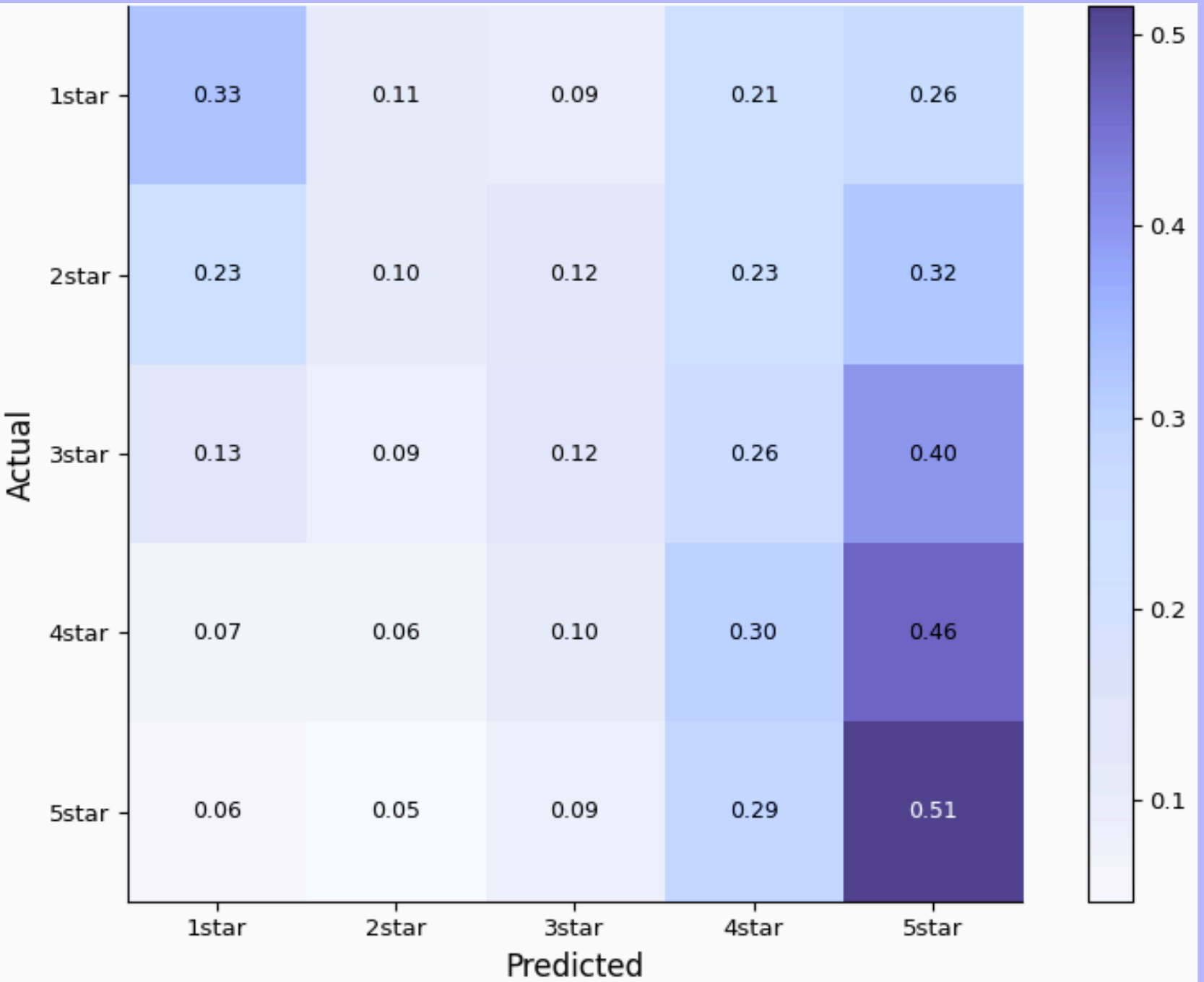
42.4%

MACRO F1

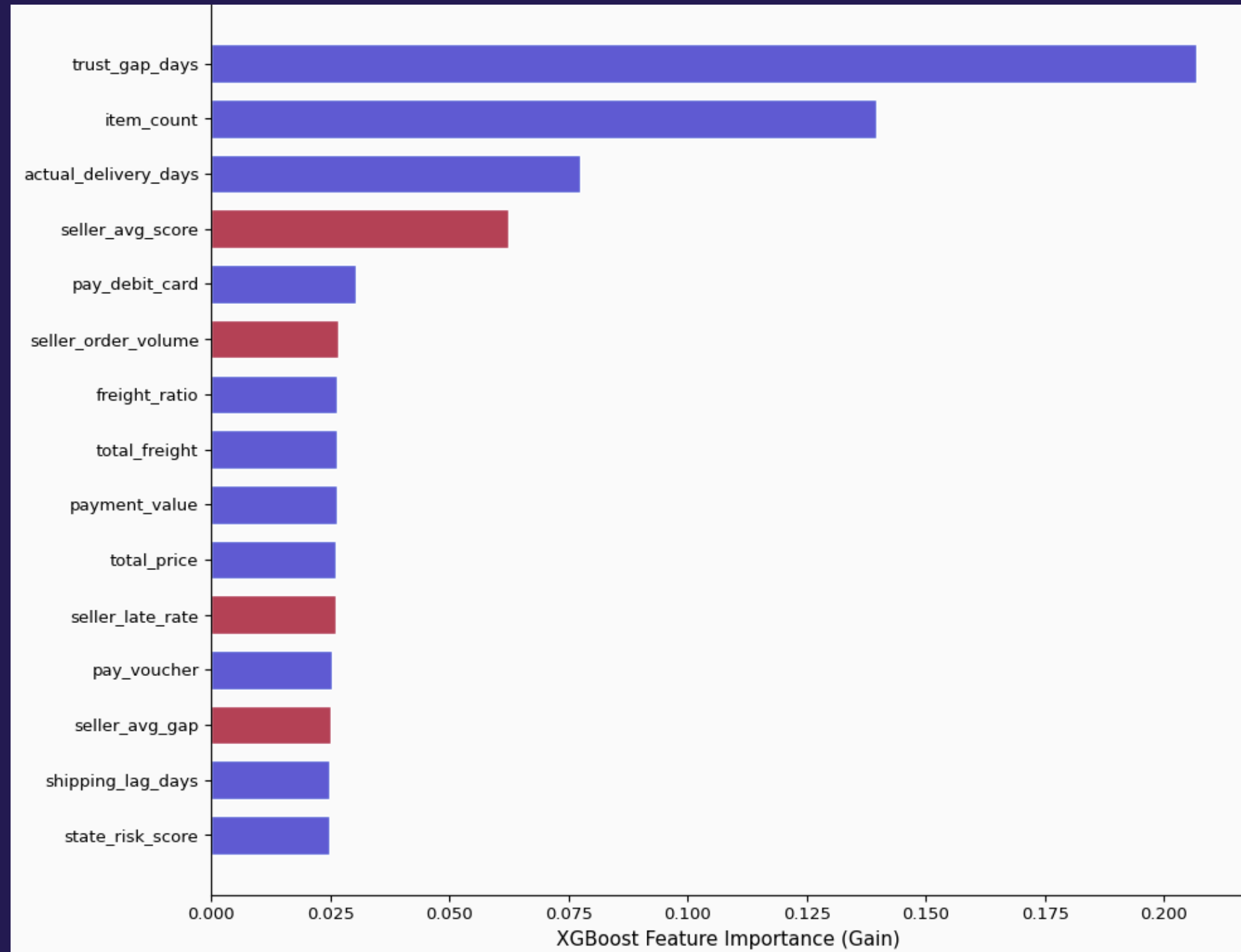
0.26

CONFUSION MATRIX

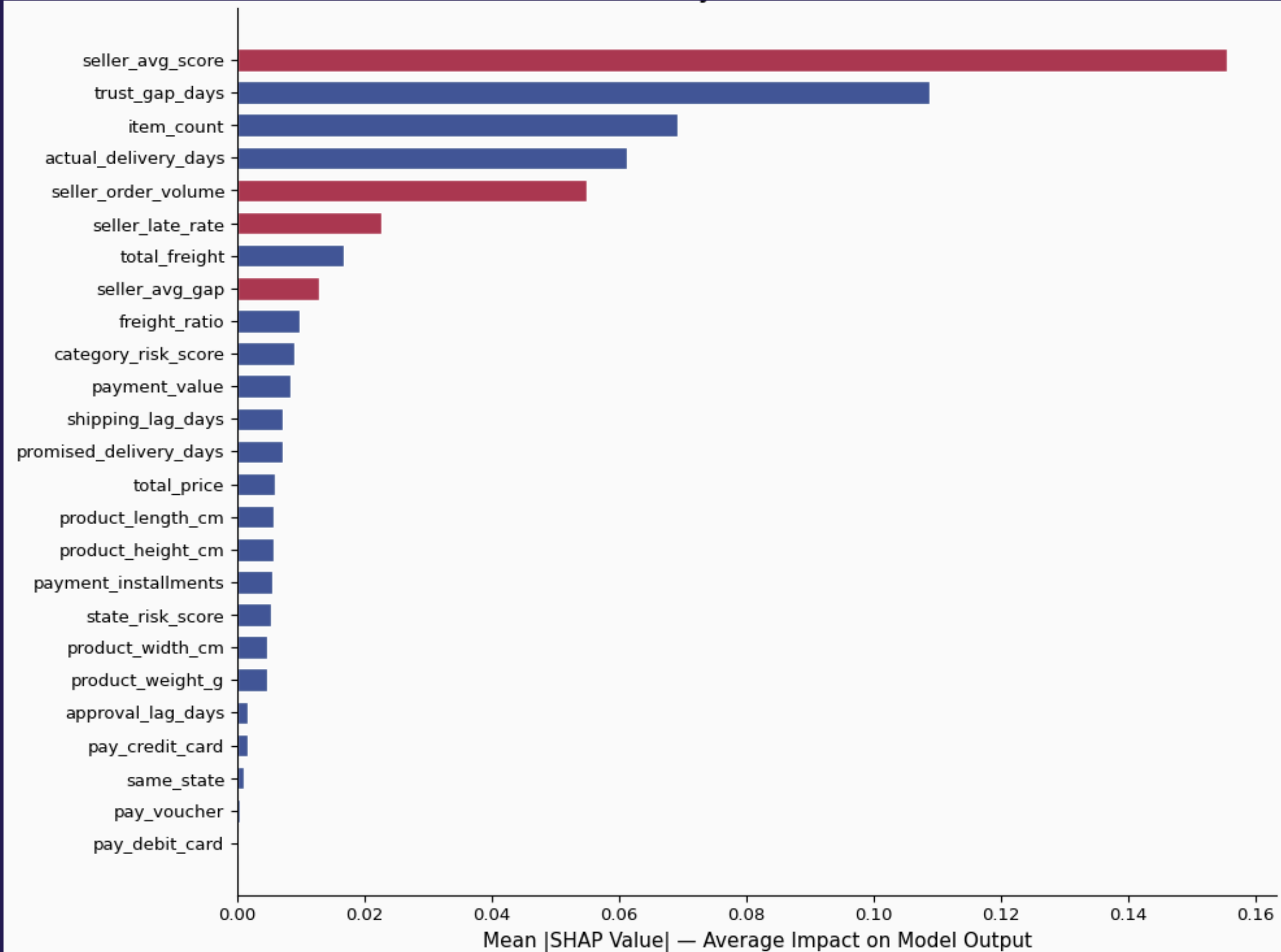
	PRECISION	RECALL	F1
1-STAR	0.26	0.33	0.29
2-STAR	0.04	0.10	0.06
3-STAR	0.09	0.12	0.11
4-STAR	0.20	0.30	0.24
5-STAR	0.70	0.51	0.59
MACRO AVG	0.26	0.27	0.26
ACCURACY			0.42



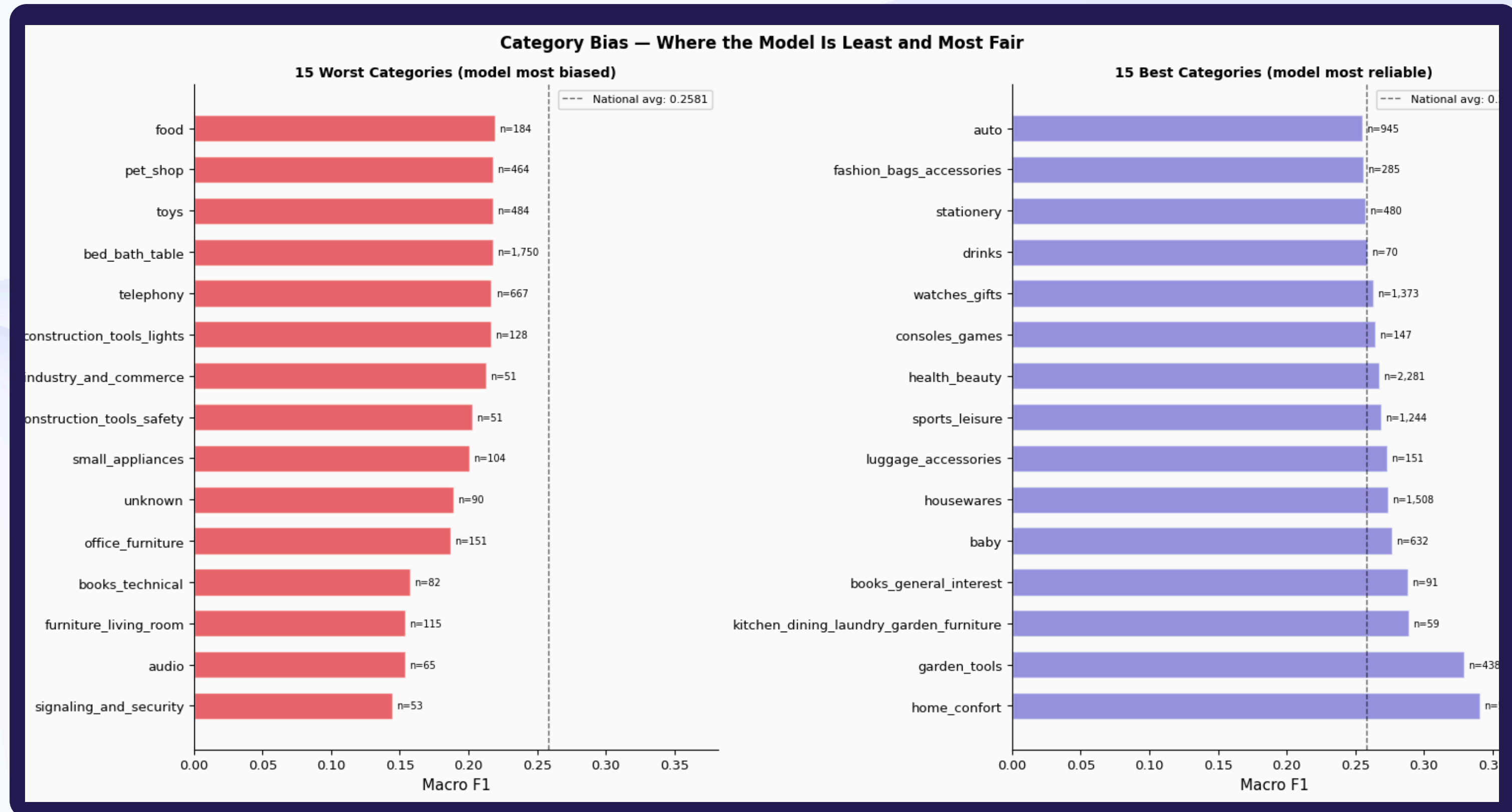
FEATURE IMPORTANCE



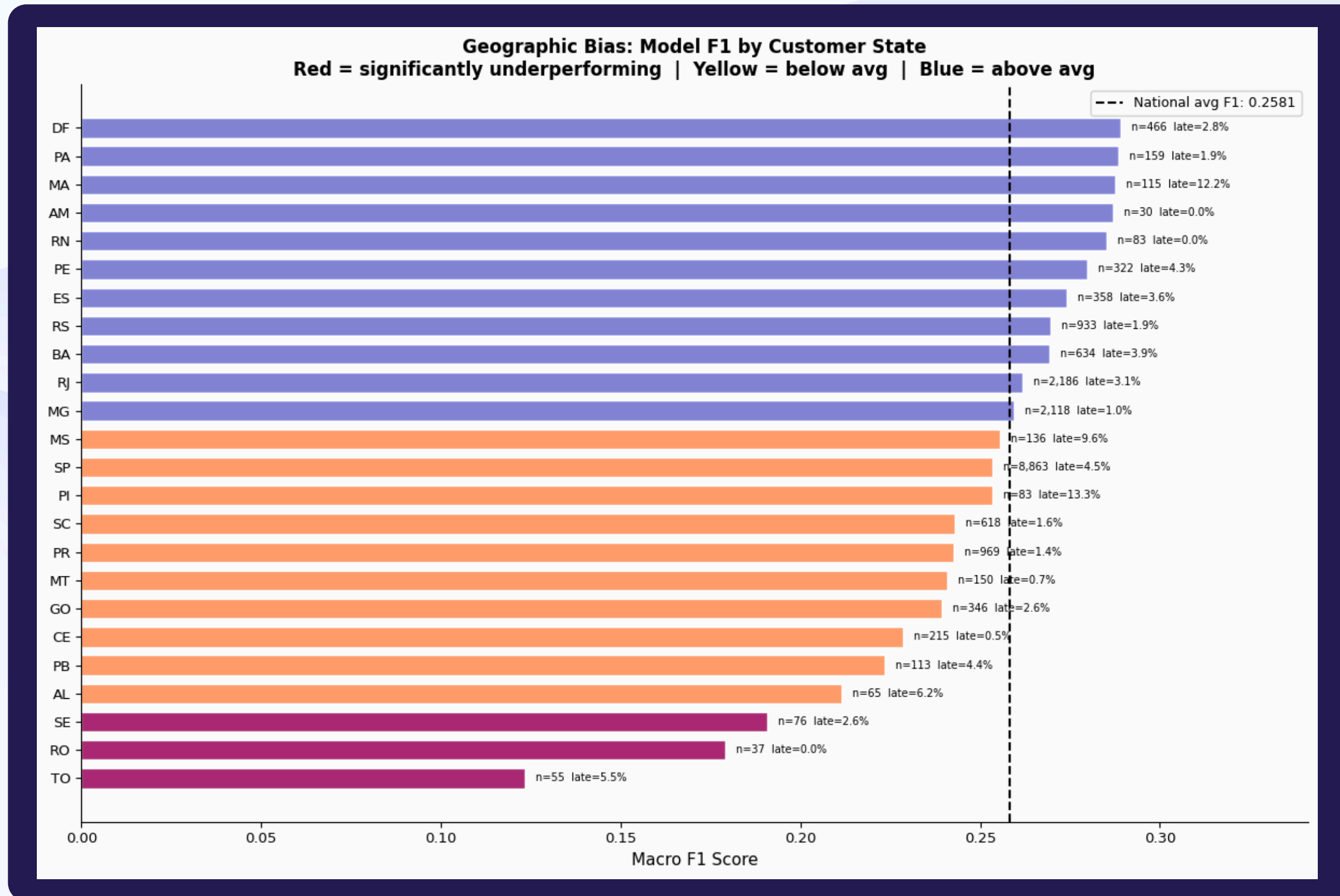
SHAP FEATURE IMPACT



BIAS ANALYSIS: CATEGORY BIAS



BIAS ANALYSIS: GEOGRAPHIC BIAS



Key Insights

- **Delivery timing alone does not explain low reviews** - a seller with a poor reputation gets bad reviews even when delivering early
- The **11.5-day buffer** Olist builds in saves most orders but late deliveries still are a problem
- **Black Friday Nov 2017** caused a review score collapse; volume spikes break delivery systems and destroy trust at scale
- **Same-state deliveries** take 6 days median vs 12 days for different-state.

Limitations & Future Works

Limitations

- Imbalanced data
- minority classes nearly unpredictable
- Dataset is 2016–2018 only: not generalizable without retraining
- No product quality, seller behavior text, or customer mood data available

Future Works

- Add NLP on review text to capture sentiment as additional signal
- Build seller risk scoring dashboard for proactive intervention
- Retrain on post-2018 data to test if Trust Gap patterns hold



**Thank
You**